

Vol 7 Chapter 3 — Electrostatics

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 3 — Electrostatics

Level 1 — one sentence

Electrostatics — Coulomb's law, electric potential, Poisson and Laplace equations, capacitors and dielectrics — is the time-independent restriction of substrate $U(1)$ gauge field equations to static sources, with $\vec{E} = -\nabla\phi$ and the scalar potential ϕ satisfying Poisson's equation $\nabla^2\phi = -\rho/\epsilon_0$, all derivable from the substrate's $SO(2)$ -charge K-type structure.

Level 2 — graduate-physicist precision

3.1 Coulomb's law

The force between two static point charges q_1, q_2 separated by $\vec{r}$:

$$\vec{F} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \hat{r}$$

Attractive for opposite signs; repulsive for same signs. The proportionality constant $1/(4\pi\epsilon_0) \approx 9 \times 10^9 \text{ N m}^2/\text{C}^2$ is Coulomb's constant.

From a single charge, the electric field at distance r :

$$\vec{E}(\vec{r}) = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

Force on test charge q_t at that location: $\vec{F} = q_t \vec{E}$.

Substrate-mechanism: the static substrate $U(1)$ source ($J^0 = c\rho$ with ρ a point charge) produces a static $A^0 = \phi/c$ scalar potential via Maxwell's source equation in the static limit, which becomes Poisson's equation (Section 3.3 below). The resulting field is the Coulomb $1/r^2$.

3.2 The electric potential

For static fields, $\nabla \times \vec{E} = 0$ (Faraday with $\partial_t \vec{B} = 0$). This means $\vec{E}$ is a gradient:

$$\vec{E} = -\nabla\phi$$

with ϕ the electric scalar potential (volts). Potential is defined up to additive constant; only differences are physical.

For a point charge q at origin: $\phi(r) = q/(4\pi\epsilon_0 r)$. Falls off as $1/r$ (vs $1/r^2$ for $\vec{E}$).

For a distribution $\rho(\vec{r})$:

$$\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3r'$$

(Green's function for the Laplacian).

3.3 Poisson and Laplace equations

Taking $\nabla \cdot$ of $\vec{E} = -\nabla\phi$ and using Gauss:

$$\nabla \cdot \vec{E} = -\nabla^2\phi = \rho/\epsilon_0$$

So:

$$\nabla^2\phi = -\rho/\epsilon_0 \quad (\text{Poisson's equation})$$

In source-free regions: $\nabla^2\phi = 0$ (Laplace's equation). Solutions to Laplace are harmonic functions — their values at any point equal the average over surrounding spheres (mean-value property).

Boundary value problems for electrostatics: given a charge distribution + boundary conditions (e.g., conductors at specified potentials), solve Laplace/Poisson. Standard methods: image charges, separation of variables, multipole expansion, Green's functions.

3.4 Electric flux and Gauss's law applications

Gauss's law: $\oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}/\epsilon_0$.

Useful for problems with symmetry: - Point charge: spherical Gaussian surface $\rightarrow \vec{E} = q/(4\pi\epsilon_0 r^2)\hat{r}$ - Infinite line charge: cylindrical surface $\rightarrow \vec{E} = \lambda/(2\pi\epsilon_0 r)\hat{r}$ - Infinite plane: pillbox surface $\rightarrow \vec{E} = \sigma/(2\epsilon_0)\hat{n}$ - Spherical shell: $\vec{E} = 0$ inside; Coulomb outside

3.5 Conductors in electrostatics

For a conductor in electrostatic equilibrium: - $\vec{E} = 0$ inside conductor (any field would drive currents) - Conductor surface is equipotential ($\nabla\phi = 0$ tangentially) - All excess charge resides on the surface - $\vec{E}$ just outside surface is σ/ϵ_0 perpendicular to surface

The method of images uses these properties: e.g., a point charge above an infinite grounded plane is solved by placing an image charge of opposite sign at the mirror position.

3.6 Capacitors

A capacitor stores charge $\pm Q$ on two conductors at potential difference V . The capacitance:

$$C = Q/V$$

Parallel-plate capacitor (area A , separation d): $C = \epsilon_0 A/d$. Energy stored: $U = (1/2)Q^2/C = (1/2)CV^2$.

Substrate reading: a capacitor's stored charge is the substrate's $SO(2)$ -charged K-type accumulation on the conductor surfaces; the stored energy is the substrate's potential energy in the assembled K-type configuration.

3.7 Dielectrics and polarization

Insulators (dielectrics) have bound charges that respond to applied $\vec{E}$. The polarization $\vec{P}$ (dipole moment per unit volume) satisfies $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ where $\vec{D}$ is the displacement field.

For linear dielectrics: $\vec{P} = \epsilon_0 \chi_e \vec{E}$ with χ_e the susceptibility, $\epsilon = \epsilon_0(1 + \chi_e)$ the permittivity.

Modified Gauss's law: $\nabla \cdot \vec{D} = \rho_{\text{free}}$.

Substrate-mechanism reading: dielectric polarization is the substrate's atomic K-type response to applied $U(1)$ field; bound charges are substrate K-type rearrangements within atoms.

3.8 Energy and force

Energy density in the electric field: $u_E = (1/2)\epsilon_0 E^2$.

Force on a charge distribution: $\vec{F} = \int \rho \vec{E} d^3r$ (direct), or computed from energy gradient $\vec{F} = -\nabla U$.

Substrate reading: field energy density is the substrate's $SO(2)$ -connection-amplitude squared, integrated over space; force on charges is the substrate's K-type-cluster gradient toward lower-energy configurations (Casey's Principle).

3.9 Worked example: parallel-plate capacitor

Two plates of area $A = 10 \text{ cm}^2 = 10^{-3} \text{ m}^2$, separation $d = 1 \text{ mm} = 10^{-3} \text{ m}$, applied voltage $V = 100 \text{ V}$.

$$C = \epsilon_0 A/d = (8.85 \times 10^{-12})(10^{-3})/(10^{-3}) = 8.85 \times 10^{-12} \text{ F} = 8.85 \text{ pF}$$

Stored charge: $Q = CV = 8.85 \times 10^{-10} \text{ C} = 0.885 \text{ nC}$.

Energy stored: $U = (1/2)CV^2 = (1/2)(8.85 \times 10^{-12})(10^4) = 4.4 \times 10^{-8} \text{ J}$.

Field between plates: $E = V/d = 10^5 \text{ V/m}$.

Substrate reading: the 0.885 nC of charge corresponds to $\sim 5.5 \times 10^9$ electrons rearranged onto the plate surfaces. Each electron is one substrate Pin(2) half-integer-weight K-type with $SO(2)$ charge weight; the capacitor's stored energy is the substrate's K-type-cluster potential.

3.10 K-audit anchors

- Chapter 2: Maxwell's equations from substrate $U(1)$ (foundation)
- T2470 (Lyra Friday May 22, 2026): substrate $SO(2)$ charge structure
- Volume 5 Chapter 9: Pin(2) particle-statistics (electron as fermion)

Level 3 — 5th-grader accessibility

Electrostatics is electricity-not-moving. Coulomb's law: two charges q_1, q_2 apart by distance r push each other with force $q_1 q_2 / (4\pi\epsilon_0 r^2)$ — opposite signs attract, same signs repel. Field lines go from $+$ to $-$. Electric potential is voltage; the field points “downhill” in voltage. Poisson's equation $\nabla^2 \phi = -\rho/\epsilon_0$ tells you the voltage everywhere given the charges. Capacitors store charge on plates at fixed voltage; capacitance $C = Q/V$, energy $\frac{1}{2}CV^2$. Conductors have zero field inside in equilibrium (charges arrange to cancel any internal field). All of this comes from Maxwell's equations restricted to time-independent fields — and Maxwell comes from the substrate's $U(1)$ gauge structure (Chapter 2).

What comes next

Chapter 4 develops magnetostatics — Ampère's law, Biot-Savart, magnetic vector potential.

Where to look this up

- Standard electrostatics: Griffiths Ch 2-4; Jackson Ch 1-3
- Method of images: Griffiths Ch 3
- Capacitors and dielectrics: Jackson Ch 4
- BST anchors: Chapter 2 Maxwell; T2470 SO(2) charge
- Volume 5 Chapter 9: spin-statistics (electron as fermion)